Tournament Dynamics Beyond ICM

Chapter Overview

Chosen theme: Decision-making under nonlinear utility in tournament settings, where chip value depends on stack size and payout structure rather than being constant.

Central question: What general principles govern optimal tournament decisions when chip value is nonlinear?

Connection to poker: In poker tournaments, chips do not translate linearly into winnings because payouts are discrete and top-heavy. As a result, players must make decisions based on how chips affect their probability of finishing in different payout positions, not just their raw chip count. This leads to risk-averse behavior, changing all-in thresholds, and strategic adjustments near the bubble or final table. Understanding nonlinear chip value explains why strong players sometimes pass on profitable chip gambles.

Connection to earlier chapters: This chapter builds on earlier ideas of expected value and decision-making under uncertainty, extending them from linear chip EV to utility-based reasoning. Concepts like hand equity, pot odds, and optimal play still apply, but they must now be interpreted through a nonlinear objective function. The chapter also connects to prior discussions of risk and variance, formalizing them through concave utility functions and marginal value of chips.

Core mathematical tools:

- Concave utility functions and diminishing marginal utility
- Expected utility maximization (replacing chip EV)
- Marginal value of chips and derivatives
- Risk premium and certainty equivalents
- Bubble factor as a ratio of marginal utilities
- Modeling tournament payouts as nonlinear mappings from chip stacks to outcomes

AI Collaboration Log

Keep a brief record of how your group used AI. This does not need to be long, but it should be honest and specific.

Task	How AI Was Used	What the Group Revised or Verified
Example: definitions	Asked AI to draft 3 candidate definitions	Revised wording and checked against chapter notation
Running example	Generated structured example	Verified calculations
Theorem	Drafted concavity proof	Checked correctness

Motivating Example

You are deep in a poker tournament. The blinds are increasing, and every decision carries significant weight. You have 18 big blinds remaining—enough to still maneuver, but not enough to play passively.

You are dealt Ace-Queen offsuit in middle position. You raise, and only the big blind calls. The flop comes:

Q 7 2

You have top pair with a strong kicker. The pot is 6 big blinds.

Your opponent checks.

At first glance, this seems straightforward—you likely have the best hand. But the situation is more complicated than it appears.

If you bet and your opponent folds, you win the pot immediately. If they call, you must consider how your hand performs against their possible range. If they check-raise all-in, you are suddenly facing a tournament-defining decision: call and risk elimination, or fold and preserve your remaining chips.

Suppose your opponent shoves all-in for your remaining 15 big blinds after your bet. The pot now offers you favorable odds—but only if your hand has enough equity against their range.

Now the key question emerges:

How strong does your hand need to justify calling?

Answering this is not about intuition alone. It requires translating the situation into mathematics: - What are the pot odds you are being offered? - How do you quantify your probability of winning (your equity)? - How do these quantities determine whether calling is profitable?

Even more subtly, this decision is not just about chips—it is about survival in a tournament. Losing all your chips has a different consequence than losing the same number in a cash game.

This example reveals a deeper mathematical problem at the heart of poker:

How can we model decision-making under uncertainty, where outcomes depend on both probability and strategic interaction?

To answer this, we need tools from probability, expected value, and game theory. These tools will allow us to move beyond intuition and make decisions that are mathematically justified—even in high-pressure, uncertain situations.

Mathematical Framework and Poker Theory

Definitions and Setup

We model a tournament with n players and a fixed payout structure determined by finishing position. Let
- $c_i \in \mathbb{R}_+$ denote the chip stack of player i,
- $\mathbf{c} = (c_1, \dots, c_n)$ denote the full vector of chip stacks,
- $\mathbf{c}_{-i}$ denote the vector of all stacks except player i's.

Let

$$U_i(\mathbf{c})$$

denote the **tournament utility** of player i, defined as their expected monetary payout given the current stack configuration.

Unlike in cash games, this utility depends on relative position rather than absolute chip counts.

Types of Utility Functions

- Linear: $U_i(c_i) = c_i$
- Payout-based (ICM-style)

- Future game simulation
- **Skill-weighted utility (used in this chapter)**

Skill-Weighted Utility Model

We define skill weights $s_i > 0$ and model:

$$P_i(1\text{st}) = \frac{s_i c_i}{\sum_j s_j c_j}$$

$$U_i(\mathbf{c}) = \sum_k P_i(k) \cdot \text{Payout}_k$$

Running Example

We use a consistent example throughout the chapter.

- 3 players
- Payouts: $(100, 60, 0)$
- Stacks:

$$\mathbf{c} = (40, 35, 25)$$

- Skill weights:

$$s_1 = 1.3, \quad s_2 = 1.0, \quad s_3 = 0.9$$

Compute:

$$\sum s_j c_j = 109.5$$

$$P_1(1\text{st}) \approx 0.475$$

Approximate:

$$P_1(2\text{nd}) \approx 0.35$$

Thus:

$$U_1(\mathbf{c}) \approx 68.5$$

Definition 1 (Nonlinear Utility Function)

Definition. A nonlinear utility function is a function

$$U_i : \mathbb{R}_+^n \to \mathbb{R}$$

such that the change in utility from gaining chips depends on the current stack size and the stacks of other players.

Formally, for $\Delta > 0$,

$$U_i(c_i + \Delta, \mathbf{c}_{-i}) - U_i(c_i, \mathbf{c}_{-i})$$

is not constant in c_i.

Explanation of Components

- $U_i(\mathbf{c})$: expected payout of player i
- c_i: player i's chip stack
- $\mathbf{c}_{-i}$: opponents' stacks
- Δ: change in chips

Interpretation

In a linear system (such as a cash game), each chip has constant value:

$$U_i(c_i + \Delta) - U_i(c_i) = \text{constant}.$$

In tournaments, this is not true. For example: - Increasing from a short stack significantly improves survival probability
- Increasing an already large stack produces only a small gain in expected payout

This implies diminishing marginal returns, typically expressed as

$$\frac{\partial^2 U_i}{\partial c_i^2} < 0.$$

Definition 2 (Risk Premium)

Definition. Let X be a random variable representing a change in player i's chip stack. The risk premium π is defined implicitly by

$$U_i(\mathbb{E}[c_i + X] - \pi, \mathbf{c}_{-i}) = \mathbb{E}[U_i(c_i + X, \mathbf{c}_{-i})].$$

Explanation of Components

- X: random chip outcome (e.g., $+\Delta$ or $-\Delta$)
- $\mathbb{E}[c_i + X]$: expected chip stack after the gamble
- $\mathbb{E}[U_i(c_i + X, \mathbf{c}_{-i})]$: expected tournament utility
- π: amount of chips the player would give up to avoid risk

Interpretation

The risk premium measures how much a player dislikes uncertainty.

- If $\pi = 0$, the player is risk-neutral
- If $\pi > 0$, the player is risk-averse

In tournaments, risk aversion arises because losing chips can lead to elimination, while gaining chips often provides limited additional benefit.

Definition 3 (Bubble Factor)

Definition. For $\Delta > 0$, the bubble factor for player i is

$$B_i(\mathbf{c}, \Delta) = \frac{U_i(c_i, \mathbf{c}_{-i}) - U_i(c_i - \Delta, \mathbf{c}_{-i})}{U_i(c_i + \Delta, \mathbf{c}_{-i}) - U_i(c_i, \mathbf{c}_{-i})}.$$

Explanation of Components

The numerator

$$U_i(c_i, \mathbf{c}_{-i}) - U_i(c_i - \Delta, \mathbf{c}_{-i})$$

represents the utility lost when chips are lost.

The denominator

$$U_i(c_i + \Delta, \mathbf{c}_{-i}) - U_i(c_i, \mathbf{c}_{-i})$$

represents the utility gained when chips are won.

Interpretation

- $B_i > 1$: losing chips is more costly than gaining chips is beneficial
- $B_i = 1$: symmetric value (cash-game case)
- $B_i \gg 1$: strong risk aversion (e.g., near payout thresholds)

This ratio captures how tournament incentives distort decision-making.

Connecting the Definitions

These concepts are closely related.

- Nonlinear utility creates diminishing marginal returns
- Diminishing returns lead to risk aversion
- Risk aversion is quantified by the risk premium
- The bubble factor measures the asymmetry between gains and losses

Together, they imply that chip value depends on both stack size and tournament context.

Main Mathematical Development

[Break the chapter into subsections that develop one primary theme. Use theorem-style statements when appropriate. Include proofs when they illuminate structure or reasoning.]

Subsection A: Marginal Value of Chips

A central feature of tournament poker is that the value of an additional chip is not constant. Instead, it depends on both a player's current stack and the distribution of chips across all players.

Definition (Marginal Value of Chips)

The **marginal value of chips** for player i is defined as the rate at which tournament utility changes with respect to their chip stack:

$$\frac{\partial U_i}{\partial c_i}(\mathbf{c}).$$

This quantity measures how much a small increase in chips improves a player's expected payout.

Interpretation of the Derivative

- If $\frac{\partial U_i}{\partial c_i}$ is large, gaining chips significantly increases expected payout
- If $\frac{\partial U_i}{\partial c_i}$ is small, additional chips provide little benefit

Thus, the derivative captures the **local value of chips** at a given stack size.

Key Property: Diminishing Marginal Value

In tournament settings, utility is typically concave in c_i, meaning:

$$\frac{\partial^2 U_i}{\partial c_i^2}(\mathbf{c}) < 0.$$

This implies that: - The first chips a player gains are very valuable
- Each additional chip contributes less than the previous one

Example (Qualitative)

- A player with a very short stack gains substantial survival probability from doubling up
- A chip leader gains relatively little from increasing an already dominant stack

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Dependence on Stack Distribution

Unlike standard economic models, the marginal value of chips depends not only on c_i, but on the entire vector $\mathbf{c}$:

$$\frac{\partial U_i}{\partial c_i}(\mathbf{c}) = f(c_1, c_2, \dots, c_n).$$

This reflects the fact that tournament outcomes depend on relative chip positions.

Consequences

- If many opponents are short-stacked, survival becomes more valuable
- If stacks are evenly distributed, accumulating chips becomes more valuable
- If one player dominates, marginal gains for that player are minimal

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Discrete Approximation

Since chips change in discrete amounts, the marginal value is often approximated by:

$$U_i(c_i + \Delta, \mathbf{c}_{-i}) - U_i(c_i, \mathbf{c}_{-i}),$$

for small $\Delta > 0$.

This gives a practical way to estimate marginal value without calculus.

Relation to Risk and Bubble Factor

The marginal value of chips directly influences both risk premium and bubble factor:

- When marginal value is high (short stacks), players are incentivized to take risks
- When marginal value is low (large stacks), players can apply pressure
- When losing chips significantly reduces marginal utility, the bubble factor increases

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Result

The marginal value of chips depends on the stack distribution:

$$\frac{\partial U_i}{\partial c_i}(\mathbf{c}) \text{ varies with both } c_i \text{ and } \mathbf{c}_{-i}.$$

Implication for Strategy

Because marginal chip value is not constant:

- Decisions that maximize expected chips may not maximize expected payout
- Players must account for how chips translate into finishing probabilities
- Optimal play requires evaluating changes in utility, not just chip counts

This is the fundamental reason tournament strategy differs from cash-game strategy.

Subsection B: Tournament EV vs Chip EV

Tournament Expected Value vs. Chip Expected Value

A central distinction in tournament poker is the difference between **chip expected value** and **tournament expected value**. While chip expected value measures gains and losses in chips, tournament expected value measures changes in expected payout. Because tournament utility is nonlinear, these two quantities can lead to different strategic decisions.

Definition (Chip Expected Value)

Consider a gamble in which player i wins $\Delta > 0$ chips with probability p and loses Δ chips with probability $1 - p$.

The **chip expected value** (chip-EV) of this gamble is

$$\mathrm{EV}_{\text{chips}} = p \cdot \Delta + (1 - p)(-\Delta) = (2p - 1)\Delta.$$

Interpretation

- $\mathrm{EV}_{\mathrm{chips}} > 0$: the gamble is profitable in chips
- $\mathrm{EV}_{\mathrm{chips}} = 0$: the gamble is fair
- $\mathrm{EV}_{\mathrm{chips}} < 0$: the gamble loses chips on average

Chip-EV treats each chip as having equal value, independent of context.

Definition (Tournament Expected Value)

The **tournament expected value** (tournament-EV) evaluates the same gamble using the utility function U_i:

$$\mathrm{EV}_{\mathrm{tourn}} = p \cdot U_i(c_i + \Delta, \mathbf{c}_{-i}) + (1-p) \cdot U_i(c_i - \Delta, \mathbf{c}_{-i}) - U_i(\mathbf{c}).$$

Explanation of Components

- $U_i(c_i + \Delta, \mathbf{c}_{-i})$: utility if the gamble is won
- $U_i(c_i - \Delta, \mathbf{c}_{-i})$: utility if the gamble is lost
- $U_i(\mathbf{c})$: current utility baseline
- p: probability of winning the gamble

Interpretation

Tournament-EV measures how a decision affects **expected payout**, not chip count.

Key Difference

Chip-EV assumes linear utility:

$$U_i(c_i) = c_i,$$

while tournament-EV uses a nonlinear utility function.

Because of this,

$$\mathbb{E}[U_i(c_i + X)] \neq U_i(\mathbb{E}[c_i + X]).$$

This inequality is the mathematical source of the difference between the two frameworks.

When the Two Agree

If utility is linear, then:

$$U_i(c_i + \Delta) - U_i(c_i) = \Delta,$$

and therefore

$$\mathrm{EV}_{\mathrm{tourn}} = \mathrm{EV}_{\mathrm{chips}}.$$

This corresponds to cash-game settings.

When the Two Differ

In tournaments, utility is concave, implying risk aversion:

$$U_i(\mathbb{E}[c_i + X]) > \mathbb{E}[U_i(c_i + X)].$$

As a result, it is possible that:

$$\mathrm{EV}_{\mathrm{chips}} > 0 \quad \text{but} \quad \mathrm{EV}_{\mathrm{tourn}} < 0.$$

Interpretation

A gamble may gain chips on average but still reduce expected payout due to the risk of losing chips.

Role of the Bubble Factor

Using the bubble factor, we can rewrite the decision condition.

A player should accept the gamble only if:

$$p \cdot (U_i(c_i + \Delta) - U_i(c_i)) \geq (1 - p) \cdot (U_i(c_i) - U_i(c_i - \Delta)).$$

Rearranging gives:

$$\frac{p}{1-p} \geq B_i(\mathbf{c}, \Delta).$$

Interpretation

- The required odds depend on the bubble factor
- When $B_i > 1$, the player needs better-than-even odds to justify the gamble

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Strategic Implications

The divergence between chip-EV and tournament-EV leads to systematic differences in play:

- Players may fold positive chip-EV hands to avoid elimination
- Medium stacks avoid risk due to high bubble factors
- Large stacks can take advantage of others' risk aversion
- Short stacks may accept marginal gambles due to high marginal utility

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Insight

Maximizing chip expected value is not equivalent to maximizing tournament expected value.

Optimal tournament decisions must account for: - nonlinear utility, - risk premium, - and the asymmetry between gains and losses.

This distinction is fundamental to understanding tournament dynamics beyond simple chip-based reasoning.

Subsection C: All-In Decision Thresholds

A fundamental application of tournament utility is determining whether a player should accept an all-in gamble. This section derives a precise probability threshold that replaces chip-based intuition with utility-based reasoning.

Setup

Consider a binary gamble faced by player i:

- With probability p, player i wins $\Delta > 0$ chips
- With probability $1 - p$, player i loses Δ chips

Let $\mathbf{c}_{-i}$ denote the fixed chip stacks of all other players.

We compare tournament expected utility before and after the decision.

Tournament Decision Condition

The player should accept the gamble if and only if:

$$\mathrm{EV}_{\mathrm{tourn}} \geq 0,$$

which expands to:

$$p \cdot U_i(c_i + \Delta, \mathbf{c}_{-i}) + (1 - p) \cdot U_i(c_i - \Delta, \mathbf{c}_{-i}) \geq U_i(\mathbf{c}).$$

Rearranging terms gives:

$$p \cdot (U_i(c_i + \Delta, \mathbf{c}_{-i}) - U_i(c_i, \mathbf{c}_{-i})) \geq (1 - p) \cdot (U_i(c_i, \mathbf{c}_{-i}) - U_i(c_i - \Delta, \mathbf{c}_{-i})).$$

Probability Threshold Form

Dividing both sides yields a clean decision rule:

$$\frac{p}{1-p} \geq \frac{U_i(c_i, \mathbf{c}_{-i}) - U_i(c_i - \Delta, \mathbf{c}_{-i})}{U_i(c_i + \Delta, \mathbf{c}_{-i}) - U_i(c_i, \mathbf{c}_{-i})}.$$

Recognizing the right-hand side as the bubble factor, we obtain:

$$\frac{p}{1-p} \geq B_i(\mathbf{c}, \Delta).$$

Solving for the Critical Probability

We can express the condition directly in terms of a minimum required winning probability:

$$p \geq \frac{B_i(\mathbf{c}, \Delta)}{1 + B_i(\mathbf{c}, \Delta)}.$$

Interpretation

This inequality provides a direct bridge between tournament structure and decision-making:

- If $B_i = 1$, then $p \geq \frac{1}{2}$ (cash-game threshold)
- If $B_i > 1$, the required probability increases above $1/2$
- If $B_i \gg 1$, the player requires a very high chance of winning to justify calling

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Strategic Meaning

This result formalizes how tournament incentives distort standard pot-odds reasoning:

- Chip EV would only require $p > \frac{1}{2}$ for a fair gamble
- Tournament EV requires adjusting this threshold upward by the bubble factor
- As a result, players near payout thresholds must demand significantly better odds than in cash games

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Insight

The bubble factor acts as a multiplier on required equity:

$$\text{Required Equity} = \frac{B_i}{1 + B_i}.$$

Thus, optimal tournament decisions depend not only on pot odds, but also on stack-dependent utility curvature.

Poker Connection

The mathematical framework developed in this chapter directly explains several fundamental features of tournament poker strategy, particularly in situations where chip-based reasoning fails.

At the most basic level, poker decisions are often initially analyzed using chip expected value (chip-EV), where outcomes are measured purely in terms of chips gained or lost. However, tournament play introduces payout-dependent nonlinearities that fundamentally alter optimal decision-making.

Tournament Decisions and Stack Pressure

The concept of nonlinear utility explains why identical chip situations can lead to different optimal decisions depending on stack sizes and payout structure.

- A short-stacked player may correctly take a high-variance gamble that is slightly negative in chip-EV, because the marginal utility of survival is extremely high.
- A medium-stacked player in the same spot may fold the same hand, since the risk of elimination carries a large utility penalty (high bubble factor).
- A chip leader may take aggressive lines that are chip-EV neutral or slightly negative, because additional chips contribute relatively little to tournament utility.

Thus, optimal strategy depends on the full stack distribution $\mathbf{c}$, not just individual hand strength.

All-In Decisions as Utility Thresholds

The all-in decision framework derived earlier translates directly into real poker situations such as calling or folding against a shove.

Instead of comparing raw chip equity, players must compare:

$$\mathrm{EV}_{\mathrm{tourn}} = p \cdot U_i(c_i + \Delta, \mathbf{c}_{-i}) + (1-p) \cdot U_i(c_i - \Delta, \mathbf{c}_{-i}) - U_i(\mathbf{c}),$$

which leads to the threshold condition:

$$\frac{p}{1-p} \geq B_i(\mathbf{c}, \Delta).$$

In practice, this means that the same hand may be:

- a profitable call in chip-EV terms,
- but a fold in tournament EV terms if the bubble factor is sufficiently large.

Stack-Based Strategic Roles

The mathematical structure also explains common strategic roles observed in tournament play:

- **Short stacks** behave as high-risk optimizers, maximizing survival probability rather than chip accumulation.
- **Medium stacks** are the most constrained, as they face the highest bubble factors and must avoid confrontation with both larger and smaller stacks.
- **Large stacks** function as pressure agents, exploiting opponents' asymmetric utility by applying aggression that forces others into high-risk decisions.

These behaviors are not heuristic—they emerge directly from differences in marginal utility:

$$\frac{\partial U_i}{\partial c_i}(\mathbf{c}).$$

Key Insight

Tournament poker cannot be fully understood through chip arithmetic alone. Instead, every decision reflects changes in a nonlinear utility landscape shaped by:

- stack sizes,
- opponent distributions,
- and payout structure.

The mathematics shows that poker strategy is fundamentally about optimizing expected utility under a changing and highly state-dependent valuation of chips.

Beyond Poker

[Situate the mathematics outside of poker. This section should show that the ideas matter more broadly: in economics, biology, computer science, statistics, operations research, social choice, or another area. Do not introduce major new machinery here.]

Homework Problems

[Include a small, curated problem set. Aim for variety rather than volume. Problems should reinforce definitions, encourage interpretation, and invite reasoning.]

1. **Conceptual problem.** [Insert question.]
2. **Computational problem.** [Insert question.]
3. **Proof or derivation problem.** [Insert question.]
4. **Modeling or interpretation problem.** [Insert question.]
5. **Optional challenge problem.** [Insert question.]

Revision Checklist

Before final submission, confirm that the chapter satisfies all of the following.

- ☐ The chapter has a clear title and central question.
- ☐ The motivating example genuinely motivates the mathematics.
- ☐ Definitions appear before use.
- ☐ Notation is stable and consistent.
- ☐ The chapter develops one main mathematical theme.
- ☐ Poker is treated as a mathematical model, not as strategy advice.
- ☐ The beyond-poker section broadens the chapter without changing its core mathematics.
- ☐ The homework problems are varied and aligned with the chapter.
- ☐ All AI-generated material has been checked and revised by the group.

Submission Notes

Submit the completed `.qmd` file along with any supporting figures, code, or data files. If you used AI in a substantial way, include the collaboration log above or a short appendix describing how it shaped the chapter.

The goal is not to hide AI use. The goal is to use it well: as a tool for mathematical writing, revision, and exploration, while preserving your group's responsibility for the finished mathematics.